

Numeric Approximations of ODEs

1. Introduction

Consider an Initial Value Problem (IVP) which is to be solved in the interval $[x_0, x_m]$. We divide this interval into N parts, where $x_n = x_0 + nh$ for $n = 0, 1, 2, \dots, N$ and $h = (x_m - x_0)/N$. For each step, we need to find $y_n \approx y(x_n)$ (where $y(x_n)$ is the exact real solution).

2. One Step Method

y_{n+1} is written in terms of x_n and y_n , i.e., $y_{n+1} = y_n + h\phi(x_n, y_n; h)$.

$e_n = y(x_n) - y_n$ is called the global error.

$T_n = \frac{y(x_{n+1}) - y(x_n)}{h} - \phi(x_n, y_n, h)$ is called the truncation error.

An example would be Euler's method
consider the following Taylor's series:

$$y(x_n + h) = y(x_n) + hy'(x_n) + O(h^2) \quad (1)$$

Since $x_n + h = x_{n+1}$ and by omitting $O(h^2)$, $y'(x_n) = f(x_n, y(x_n))$, we get:

$$y_{n+1} = y_n + hf(x_n, y_n) \quad (2)$$

along with $y(x_0) = y_0$.

Given in $D = \{(x, y) \mid x_0 \leq x \leq x_m, |y - y_0| \leq \epsilon\}$. If we have $|\phi(x, u, h) - \phi(x, v, h)| \leq L_\phi |u - v|$, the relation between global and truncation error is given by $|y_n - y_0| \leq \epsilon \quad \forall n = 1, 2, \dots, N$:

$$|e_n| \leq \frac{T}{L_\phi} (e^{L_\phi(x_n - x_0)} - 1) \quad (3)$$

Where $T = \max |T_n|$.

This can be used to estimate step size, for example if $y' = \tan^{-1}(y)$, $y(0) = y_0$. Let $|e_n| \leq \frac{1}{\omega} T(e^{x_n} - 1)h \leq \text{Tolerance}$:

$$h \leq \left[\frac{\omega}{T(e^{x_m} - 1)} \right] \text{Tolerance} \quad (4)$$

where $\omega = \frac{4}{\pi}$

3. Consistency and Convergence

when $h \rightarrow 0$ and $N \rightarrow \infty$:

$$\lim_{h \rightarrow 0} T_n = 0 \quad (5)$$

Method is called **consistent**

Theorem: For $y'(x) = f(x, y(x))$:

$$\text{consistent} \iff \phi(x, y, 0) = f(x, y) \quad (6)$$

If the IVP has a unique solution and is consistent in the region, then $\lim_{h \rightarrow 0} y_n = y(x_n)$. i.e., the method converges as $h \rightarrow 0$ and $N \rightarrow \infty$.

Definition: If $|T_n| \leq Kh^P$, P is called the order.

4. Runge-Kutta Methods

Idea: get more accuracy by calculating the value of f between (x_n, y_n) and (x_{n+1}, y_{n+1}) . (This is less efficient than Euler's method).

Here $\phi(x_n, y_n) = af(x_n, y_n) + bf(x_n + \alpha h, y_n + \beta hf(x_n, y_n))$. $a + b = 1$ (consistency). In order to have 2nd order accuracy, $\alpha = \beta = \frac{1}{2}$. (These are obtained by doing a Taylor expansion up to 3rd order in the expression for T_n and equating coefficients).

A Fourth Order Method:

$$y_{n+1} = y_n + \frac{1}{6}h(k_1 + 2k_2 + 2k_3 + k_4) \quad (7)$$

Where:

$$\begin{aligned} k_1 &= f(x_n, y_n) \\ k_2 &= f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}hk_1\right) \\ k_3 &= f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}hk_2\right) \\ k_4 &= f(x_n + h, y_n + hk_3) \end{aligned}$$

5. Linear Multistep Method

Idea: consider scheme

$$y_{n+1} = y_{n-1} + \frac{1}{3}h[f(x_{n-1}, y_{n-1}) + 4f(x_n, y_n) + f(x_{n+1}, y_{n+1})]$$

here to evaluate we need y_{n-1} and y_n In general a linear k -step method is

$$\sum_{j=0}^k \alpha_j y_{n+j} = h \sum_{j=0}^k \beta_j f(x_{n+j}, y_{n+j})$$

When is obtained only from previous values then method is called explicit
If not then its called implicit

Some examples Implicit Euler method:

$$y_{n+1} = y_n + hf_{n+1}$$

Adam S- Moulton method:

$$y_{n+3} = y_{n+2} + \frac{1}{24}h(9f_{n+3} + 19f_{n+2} - 5f_{n+1} + f_n)$$

Note: we need to find $y_0, \dots, y_{k-1}$ So these are evaluated using One-step method

Here,

$$T_n = \sum_{j=0}^k [\alpha_j y(x_{n+j}) + \beta_j h f(x_{n+j-1}, y(x_{n+j}))] / h \sum_{j=0}^k \beta_j$$

6. Zero-Stability

Tiny errors initially shouldn't add up and blow up

$$\max |y_n - z_n| \leq K \max |y_0 - z_0|$$

Where y_n and z_n are generated by same formula but different initial starting values. Tiny change in initial should only result in tiny change in calculation

7. Dahlquist's Theorems

Equivalence Theorem: States (vaguely) that if a linear multistep method is consistent ($T_n \rightarrow 0$ as $h \rightarrow 0$ and $n \rightarrow \infty$) and is zero stable (tiny initial error doesn't cause blowup) then the method converges as $h \rightarrow 0$ and $N \rightarrow \infty$

Barrier theorem: order of accuracy of a Stable k -step method cannot exceed $k + 1$ if k is odd or $k + 2$ if k is even.

Say we have a 5 step method, we have 12 parameters (α_j s and β_j s) of which 11 are independent. Since we now have 11 variable we can make taylor series term Zero to get 11 equations (and be able to Solve for α_j s and β_j s) which should be able to give 10 order. BUT!! the theorem says that if we do that then it won't be stable.

8. Stiff System (Basic Idea)

Consider

$$y' = \lambda y \quad y(0) = y_0$$

If $\lambda < 0$ this is a decreasing function So our method Should also decrease Applying euler method we get,

$$y_n = (1 + h\lambda)^n y_0$$

Therefore $|1 + h\lambda| < 1$ or $h|\lambda| < 2$, there is some constraint for the step size For System of ODES

$$y' = Ay$$

if A has all negative eigenvalues and is Significant in Size then similar to above part step size will be a problem. Hence it is called a **stiff system**

9. Absolute Stability and A-Stable

Basic Idea: for $y' = \lambda y$, $y(0) = y_0$ and $\lambda < 0$ if we apply Rk4 method we have y_{n+1} depending on y_n and some function of $h\lambda$. If this function is less than one then error shrinks and method can be called **absolute stable**.

Note that λ can be complex hence we want region in complex plane. Where the function of $h\lambda$ is less than 1 is the entire left half of complex plane then our method is **A stable**, step size doesn't affect our method's output.

This gives rise to implicit Rk4 methods (Though the computation effort makes its use limited) as explicit Rk4 methods have very small region of Stability.

10. Conclusion

While approximating the ODE we should be careful with the errors that come with it. When a one step - method is consistent we should be able to get good approximations WITH PROPER STEP SIZE! With multistep method Since we are adding a lot of steps to find the n th value a tiny initial error is being added up a lot of time which will deviate the answer by a lot in the end that's why on top of consistency we are also concerned about **Zero - Stability** (One - Step is trivially zero- stable) Further we are also concerned about **absolute stability**, Stability for fixed value of $h\lambda$.